Rigorous Examination Solutions: Set Theory and Topology

Complete Proofs and Justifications (Questions 1–5)

Question 1

Problem Statement. Let A be a set and $\mathscr{P}(A)$ be its power set. Prove that there is no injective map $f : \mathscr{P}(A) \to A$ and no surjective map $g : A \to \mathscr{P}(A)$.

Proof. We establish both claims rigorously using Cantor's diagonalization argument, and we also show how the non-existence of an injection follows directly from the non-existence of a surjection.

Part I: Non-existence of a surjective map $g : A \to \mathscr{P}(A)$. Suppose, for the sake of contradiction, that there exists a surjective function $g : A \to \mathscr{P}(A)$. Define the diagonal subset $B \subseteq A$ by

$$B := \{x \in A \mid x \notin g(x)\}.$$

Note that this construction is well-defined for every set A (including $A = \emptyset$, in which case $\mathscr{P}(\emptyset) = \{\emptyset\} \neq \emptyset$, so no map $g : \emptyset \to \mathscr{P}(\emptyset)$ exists at all).

Since $B \subseteq A$, we have $B \in \mathscr{P}(A)$. Because g is assumed to be surjective, there exists at least one element $x_0 \in A$ such that

$$g(x_0) = B.$$

By the law of excluded middle, either $x_0 \in B$ or $x_0 \notin B$. We examine both cases:

(a) **Case 1 ($x_0 \in B$):** By the defining property of B, an element $x \in A$ belongs to B if and only if $x \notin g(x)$. Consequently, $x_0 \in B \implies x_0 \notin g(x_0)$. However, since $g(x_0) = B$, this implies $x_0 \notin B$, contradicting our assumption that $x_0 \in B$.

(b) **Case 2 ($x_0 \notin B$):** Since $g(x_0) = B$, the assumption $x_0 \notin B$ means $x_0 \notin g(x_0)$. By the definition of B, any element of A satisfying $x_0 \notin g(x_0)$ must belong to B; hence $x_0 \in B$, contradicting our assumption that $x_0 \notin B$.

In both cases we arrive at the logical contradiction $x_0 \in B \iff x_0 \notin B$. Therefore, our initial assumption is false, and no surjective map $g : A \to \mathscr{P}(A)$ exists.

Part II: Non-existence of an injective map $f : \mathscr{P}(A) \to A$. We provide two independent proofs of this result.

Method 1 (Direct Diagonal Proof). Suppose, for the sake of contradiction, that there exists an injective map $f : \mathscr{P}(A) \to A$. Consider the subset $C \subseteq A$ defined by

$$C := \{f(S) \in A \mid S \in \mathscr{P}(A) \text{ and } f(S) \notin S\}.$$

Since $C \subseteq A$, we have $C \in \mathscr{P}(A)$. Let $c := f(C) \in A$. We ask whether $c \in C$ or $c \notin C$:

(a) **Case 1 ($c \in C$):** By definition of C, there exists some subset $S \in \mathscr{P}(A)$ such that $c = f(S)$ and $f(S) \notin S$. Since $f(C) = c = f(S)$ and f is injective, it follows that $C = S$. Substituting $S = C$ into $f(S) \notin S$ yields $c = f(C) \notin C$, a contradiction.

(b) **Case 2 ($c \notin C$):** Since $c = f(C)$, the statement $c \notin C$ is equivalent to $f(C) \notin C$. Taking $S = C \in \mathscr{P}(A)$ in the defining condition for C, we see that $f(C)$ satisfies the membership criterion for C, which implies $c = f(C) \in C$, a contradiction.

Thus, in both cases we obtain a contradiction. Hence, no injective map $f : \mathscr{P}(A) \to A$ can exist.

Method 2 (Reduction to Part I). Suppose $f : \mathscr{P}(A) \to A$ is injective. Since $\mathscr{P}(A)$ is always non-empty (as $\emptyset \in \mathscr{P}(A)$), its image $f(\mathscr{P}(A)) \subseteq A$ is non-empty, which implies $A \neq \emptyset$. Define a map $g : A \to \mathscr{P}(A)$ by

$$g(a) := \begin{cases} S & \text{if } a = f(S) \text{ for some } S \in \mathscr{P}(A), \\ \emptyset & \text{if } a \notin f(\mathscr{P}(A)). \end{cases}$$

Because f is injective, for each $a \in f(\mathscr{P}(A))$ the preimage $S \in \mathscr{P}(A)$ with $f(S) = a$ is unique; thus g is a well-defined function. Moreover, for every $S \in \mathscr{P}(A)$, we have $g(f(S)) = S$, showing that $g : A \to \mathscr{P}(A)$ is surjective. This directly contradicts Part I. Therefore, no injective map $f : \mathscr{P}(A) \to A$ exists. □

Question 2

Problem Statement. Let X be an infinite set. Is the collection $\tau_\infty := \{U \subseteq X \mid X \setminus U \text{ is infinite or } U = \emptyset\}$ a topology on X? Justify your answer.

Answer. No, τ_∞ is **not** a topology on X. It violates both the first axiom ($X \in \tau_\infty$) and the second axiom (closure under arbitrary unions; in fact, it even fails closure under finite unions).

Justification. For τ_∞ to be a topology on X, it must satisfy the three standard topological axioms:

(T1) $\emptyset \in \tau_\infty$ and $X \in \tau_\infty$;

(T2) τ_∞ is closed under arbitrary unions;

(T3) τ_∞ is closed under finite intersections.

We examine each axiom in turn:

1. Violation of Axiom (T1) ($X \notin \tau_\infty$). By explicit definition, $\emptyset \in \tau_\infty$. However, consider the whole space $U = X$. Since X is an infinite set, $X \neq \emptyset$. Furthermore, the complement of X in X is

$$X \setminus X = \emptyset,$$

which has cardinality 0 and is therefore a **finite** set, not an infinite set. Because $U = X$ neither equals $\emptyset$ nor has an infinite complement in X, we have

$$X \notin \tau_\infty.$$

Thus, Axiom (T1) is violated.

2. Violation of Axiom (T2) (Failure of closure under unions). Even if one were to adjoin X to the collection, τ_∞ fails to be closed under arbitrary unions—and even fails under the union of two sets:

- **Counterexample via arbitrary unions of singletons:** For any element $x \in X$, consider the singleton $U_x = \{x\}$. Since X is infinite, removing a single element leaves an infinite set, so $X \setminus \{x\}$ is infinite and $\{x\} \in \tau_\infty$ for every $x \in X$. Fix a point $x_0 \in X$ and consider the union
$$U := \bigcup_{x \in X \setminus \{x_0\}} \{x\} = X \setminus \{x_0\}.$$
Since X is infinite, $U = X \setminus \{x_0\}$ is non-empty (and $U \neq X$). Yet its complement is $X \setminus U = \{x_0\}$, which is a finite set of cardinality 1. Therefore, $U \notin \tau_\infty$.
- **Counterexample via a finite union of two sets:** Since X is infinite, X contains a countably infinite subset $C = \{x_1, x_2, x_3, \dots\}$ with $x_i \neq x_j$ for $i \neq j$. Partition C into the set of even-indexed elements $E = \{x_2, x_4, x_6, \dots\}$ and odd-indexed elements $O = \{x_1, x_3, x_5, \dots\}$, both of which are infinite. Define
$$U_1 := X \setminus (E \cup \{x_1\}) \quad \text{and} \quad U_2 := X \setminus O.$$
Then $X \setminus U_1 = E \cup \{x_1\}$ is infinite, so $U_1 \in \tau_\infty$; similarly, $X \setminus U_2 = O$ is infinite, so $U_2 \in \tau_\infty$. However, by De Morgan's laws,
$$U_1 \cup U_2 = X \setminus \big((E \cup \{x_1\}) \cap O\big) = X \setminus \{x_1\}.$$
Since $U_1 \cup U_2 = X \setminus \{x_1\} \neq \emptyset$ and its complement $X \setminus (U_1 \cup U_2) = \{x_1\}$ is finite (of size 1), we have $U_1 \cup U_2 \notin \tau_\infty$. (Moreover, taking $V_1 = X \setminus E \in \tau_\infty$ and $V_2 = X \setminus O \in \tau_\infty$ gives $V_1 \cup V_2 = X \notin \tau_\infty$.)

3. Remark on Axiom (T3) (Closure under finite intersections holds). For completeness, note that if $U_1, U_2 \in \tau_\infty$, then either one of them is $\emptyset$ (so $U_1 \cap U_2 = \emptyset \in \tau_\infty$) or both $X \setminus U_1$ and $X \setminus U_2$ are infinite, in which case $X \setminus (U_1 \cap U_2) = (X \setminus U_1) \cup (X \setminus U_2)$ contains $X \setminus U_1$ and is therefore infinite, so $U_1 \cap U_2 \in \tau_\infty$. Nevertheless, because Axioms (T1) and (T2) fail, τ_∞ is not a topology on X. □

Question 3

Problem Statement. Let $X = \{a, b, c\}$. Determine which of the following collections of subsets form a valid topology on X. For those that fail, explicitly state which axiom or axioms are violated:

(i) $\mathcal{T}_1 = \{\emptyset, \{a\}, \{b\}, X\}$;

(ii) $\mathcal{T}_2 = \{\emptyset, \{a\}, \{a, b\}, X\}$;

(iii) $\mathcal{T}_3 = \{\emptyset, \{a, b\}, \{b, c\}, X\}$.

Solution and Verification. Recall that a collection $\mathcal{T} \subseteq \mathscr{P}(X)$ is a topology on X if and only if: (T1) $\emptyset, X \in \mathcal{T}$; (T2) $\mathcal{T}$ is closed under arbitrary unions; and (T3) $\mathcal{T}$ is closed under finite intersections.

Part (i): $\mathcal{T}_1 = \{\emptyset, \{a\}, \{b\}, X\}$**.**

- **Axiom (T1):** Satisfied, since $\emptyset \in \mathcal{T}_1$ and $X \in \mathcal{T}_1$.
- **Axiom (T2) (Arbitrary Unions): Violated.** Taking the union of $\{a\} \in \mathcal{T}_1$ and $\{b\} \in \mathcal{T}_1$ yields
$$\{a\} \cup \{b\} = \{a, b\} \notin \mathcal{T}_1.$$
- **Axiom (T3) (Finite Intersections):** Satisfied, since $\{a\} \cap \{b\} = \emptyset \in \mathcal{T}_1$, and the intersection of any element of $\mathcal{T}_1$ with $\emptyset$, X, or itself remains in $\mathcal{T}_1$.
- **Conclusion:** $\mathcal{T}_1$ is **not a valid topology** on X because it violates the **closure under unions axiom (T2)**.

Part (ii): $\mathcal{T}_2 = \{\emptyset, \{a\}, \{a, b\}, X\}$**.** Observe that the sets in $\mathcal{T}_2$ form a strictly nested chain under set inclusion:
$$\emptyset \subsetneq \{a\} \subsetneq \{a, b\} \subsetneq \{a, b, c\} = X.$$

- **Axiom (T1):** Satisfied, since $\emptyset \in \mathcal{T}_2$ and $X \in \mathcal{T}_2$.
- **Axiom (T2) (Arbitrary Unions):** Satisfied. Because $\mathcal{T}_2$ is finite and totally ordered by inclusion ($U \subseteq V$ or $V \subseteq U$ for all $U, V \in \mathcal{T}_2$), the union of any non-empty subcollection of $\mathcal{T}_2$ equals the largest element of that subcollection, which belongs to $\mathcal{T}_2$ (and the empty union is $\emptyset \in \mathcal{T}_2$). Explicitly, the only non-trivial pairwise union is $\{a\} \cup \{a, b\} = \{a, b\} \in \mathcal{T}_2$.
- **Axiom (T3) (Finite Intersections):** Satisfied. For any non-empty finite subcollection of $\mathcal{T}_2$, its intersection equals the smallest element in that subcollection, which belongs to $\mathcal{T}_2$. Explicitly, $\{a\} \cap \{a, b\} = \{a\} \in \mathcal{T}_2$.
- **Conclusion:** $\mathcal{T}_2$ **is a valid topology** on X.

Part (iii): $\mathcal{T}_3 = \{\emptyset, \{a, b\}, \{b, c\}, X\}$**.**

- **Axiom (T1):** Satisfied, since $\emptyset \in \mathcal{T}_3$ and $X \in \mathcal{T}_3$.
- **Axiom (T2) (Arbitrary Unions):** Satisfied, because $\{a, b\} \cup \{b, c\} = \{a, b, c\} = X \in \mathcal{T}_3$, and any union involving $\emptyset$, X, or a single set remains in $\mathcal{T}_3$.
- **Axiom (T3) (Finite Intersections): Violated.** Taking the intersection of $\{a, b\} \in \mathcal{T}_3$ and $\{b, c\} \in \mathcal{T}_3$ yields
$$\{a, b\} \cap \{b, c\} = \{b\} \notin \mathcal{T}_3.$$
- **Conclusion:** $\mathcal{T}_3$ is **not a valid topology** on X because it violates the **closure under finite intersections axiom (T3)**.

Collection	Axiom (T1)	Axiom (T2) (Unions)	Axiom (T3) (Intersections)	Valid Topology
(i) $\mathcal{T}_1$	Holds	Fails ($\{a\} \cup \{b\} = \{a, b\} \notin \mathcal{T}_1$)	Holds	**No**
(ii) $\mathcal{T}_2$	Holds	Holds	Holds	**Yes**
(iii) $\mathcal{T}_3$	Holds	Holds	Fails ($\{a, b\} \cap \{b, c\} = \{b\} \notin \mathcal{T}_3$)	**No**

□

Question 4

Problem Statement. Let X be a topological space and $A \subset X$. Prove that A is open in X if and only if for each $x \in A$, $\exists$ an open set U in X such that $x \in U \subseteq A$.

Proof. Let (X, τ) be a topological space and let $A \subseteq X$. We prove both directions of the biconditional statement.

Forward Direction ($\implies$). Suppose that A is open in X, so $A \in \tau$. Let $x \in A$ be arbitrary. Choose $U := A$. Since A is open in X, U is an open set in X, and by reflexivity of set inclusion ($A \subseteq A$), we immediately have

$$x \in U \subseteq A.$$

Thus, for each $x \in A$, there exists an open set U in X such that $x \in U \subseteq A$.

Reverse Direction ($\impliedby$). Conversely, suppose that for every $x \in A$, there exists an open set $U_x \in \tau$ such that $x \in U_x \subseteq A$. We claim that

$$A = \bigcup_{x \in A} U_x.$$

We establish this set equality by proving mutual inclusion:

1. **Proof that $A \subseteq \bigcup_{x \in A} U_x$:** Let $y \in A$ be arbitrary. By hypothesis, the corresponding open set U_y satisfies $y \in U_y$. Since U_y is one of the sets in the indexed family $\{U_x\}_{x \in A}$, it follows from the definition of the union that $y \in \bigcup_{x \in A} U_x$. Hence, $A \subseteq \bigcup_{x \in A} U_x$. (Note that if $A = \emptyset$, the empty union is $\emptyset$, so the inclusion holds vacuously.)
2. **Proof that $\bigcup_{x \in A} U_x \subseteq A$:** Let $y \in \bigcup_{x \in A} U_x$ be arbitrary. By definition of the union, there exists some index $x_0 \in A$ such that $y \in U_{x_0}$. By hypothesis, $U_{x_0} \subseteq A$, which implies $y \in A$. Hence, $\bigcup_{x \in A} U_x \subseteq A$.

Combining both inclusions yields $A = \bigcup_{x \in A} U_x$. Because each U_x is an open set in X and a topology τ is closed under arbitrary unions, we conclude that $A = \bigcup_{x \in A} U_x \in \tau$; that is, A is open in X. □

Question 5

Problem Statement. Let U be an open set and V be a closed set in X. Prove that $U \setminus V$ is open in X and $V \setminus U$ is closed in X.

Proof. Let (X, τ) be a topological space, let $U \subseteq X$ be an open set in X, and let $V \subseteq X$ be a closed set in X.

First, recall that for any two subsets $A, B \subseteq X$, the set difference $A \setminus B$ is given by the identity

$$A \setminus B = \{x \in X \mid x \in A \text{ and } x \notin B\} = A \cap (X \setminus B).$$

Furthermore, by the definition of a closed set in a topological space, a subset $F \subseteq X$ is closed in X if and only if its complement $X \setminus F$ is open in X.

Part I: Proof that $U \setminus V$ is open in X. Applying the set-difference identity to $U \setminus V$, we have

$$U \setminus V = U \cap (X \setminus V).$$

Since V is closed in X, its complement $X \setminus V$ is open in X by definition. By hypothesis, U is also open in X. By the finite intersection axiom of a topology, the intersection of two open sets in X is open in X. Therefore,

$$U \setminus V = U \cap (X \setminus V)$$

is open in X.

Part II: Proof that $V \setminus U$ is closed in X. We provide both a direct proof from the open-set axioms via De Morgan's laws and a dual proof via closed-set intersections:

Method 1: **(Direct proof via De Morgan's Laws)** To show that $V \setminus U$ is closed in X, it suffices by definition to prove that its complement $X \setminus (V \setminus U)$ is open in X. Using $V \setminus U = V \cap (X \setminus U)$ and De Morgan's laws, we compute:

$$\begin{aligned} X \setminus (V \setminus U) &= X \setminus \big(V \cap (X \setminus U)\big) \\ &= (X \setminus V) \cup \big(X \setminus (X \setminus U)\big) \\ &= (X \setminus V) \cup U. \end{aligned}$$

Since V is closed in X, its complement $X \setminus V$ is open in X. By hypothesis, U is also open in X. Because a topology is closed under unions, $(X \setminus V) \cup U$ is open in X. Consequently, $X \setminus (V \setminus U)$ is open in X, which by definition means that $V \setminus U$ is closed in X.

Method 2: **(Via intersections of closed sets)** Using the set-difference identity, we write

$$V \setminus U = V \cap (X \setminus U).$$

Since U is open in X and $X \setminus (X \setminus U) = U$, the set $X \setminus U$ is closed in X. We are also given that V is closed in X. Since the intersection of any two closed sets in X is closed in X, it follows immediately that $V \setminus U = V \cap (X \setminus U)$ is closed in X.

This completes the proof of both statements. □